LESSON PLAN · SCRATCH (GR 3–5) ADVANCED: ARCADE ENGINEERS

Lesson 3: Project: Starfall Catcher

Module 1 | 50 minutes | Project: clones game | Grades 3–5 · ages 9–11

Objective: Plan, build and test a catching game with clones, smooth controls, a score and an ending.

Success criteria

- My game uses clones, smooth controls and a score.
- It has a clear ending.
- I tuned it after playtesting.

Key vocabulary

- **game balance**: Making a game hard but fair.
- **parameter**: A number that changes how the game feels.
- **lives**: Chances left before the game ends.

You will need

- Projector with Kitabash (lesson video, Code Lab)
- Devices: 1 per learner or pair
- Printed worksheet (optional)
- Printed project brief

Timed lesson plan (50 min)

Time	Stage	What happens
5 min	**Warm-up (unplugged)**	Balance talk (5 min): list three numbers in the game and predict what happens if each doubles.
5 min	**Watch & plan**	Play the project video; learners plan on the project brief.
25 min	**Build**	Build-along steps, then Which part of the game handles it?. Check in halfway: everyone has something working.
7 min	**Test & peer feedback**	Swap and test each other's work: two stars and a wish.
8 min	**Improve & submit**	Improve one thing; Project: Starfall Catcher.

Unplugged warm-up

Balance talk (5 min): list three numbers in the game and predict what happens if each doubles.

Teaching the idea

The project combines clone factories, per-clone behaviour and collision, smooth input and game endings. Encourage tuning parameters after playtesting.

Analogy: A game designer is like a chef adjusting the recipe after tasting.

Live demo: Play a version that is too hard (fast, frequent) and one too easy; ask which numbers to change.

Questions to ask

Ask	Listen for
Which number makes the game harder?	Fall speed or clone frequency.
How does your game end?	Timer or lives.
What did your tester find unfair?	Specific feedback.

Differentiation

Support
Start from the Lesson 2 catch code and add only the ending.

Stretch
Good and bad clones (random costume), lives, and speed that increases over time.

Common misconceptions

Misconception	How to fix it
Endings happen automatically	You must code the time limit or lives check.
More clones = more fun	Balance matters more than quantity.

Assessment

Project rubric (also on the project brief):

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent
Clones	No clones	Clones appear but never delete	Clones fall and delete correctly	Varied clones (good/bad, random size/speed)
Controls	Hard to control	Key-press steps only	Smooth polling controls	Polished controls with limits at the edges
Score & ending	Missing	Score only	Score and a clear ending	Lives, timer and final message
Game balance	Unplayable	Too easy or too hard	Fair and fun	Gets harder over time in a fair way

Testing & reflection	Not tested	Tested alone	Playtested and tuned; explained	Several improvements explained clearly

Exit ticket
One number you tuned and why.

Homework
Play your game with someone at home and record their best score.

Answer key — Kitabash activities

1. Which part of the game handles it? (perform · categorize)

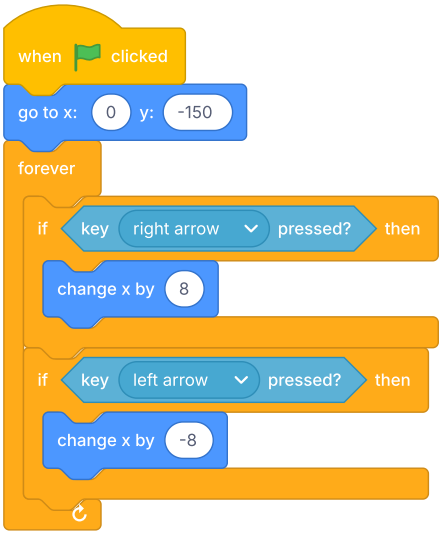
Player script: Moving with arrow keys
Original (factory): Creating a new star every 0.5 s, Hiding at the start
Clone script: Falling from a random place, Scoring when touching the player

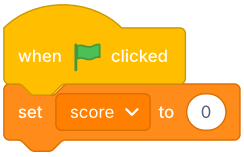
2. Project: Starfall Catcher (probe · code lab)

Goals checked: Smooth controls · Falling clones · Catching scores · Score starts at 0 · An ending

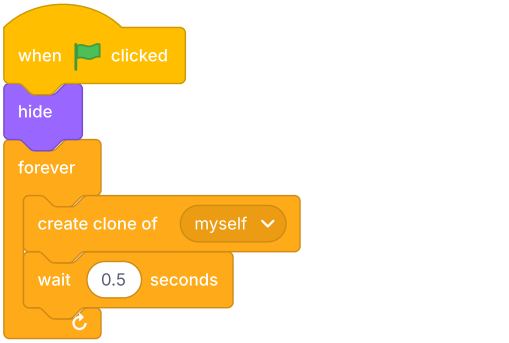
Example project:

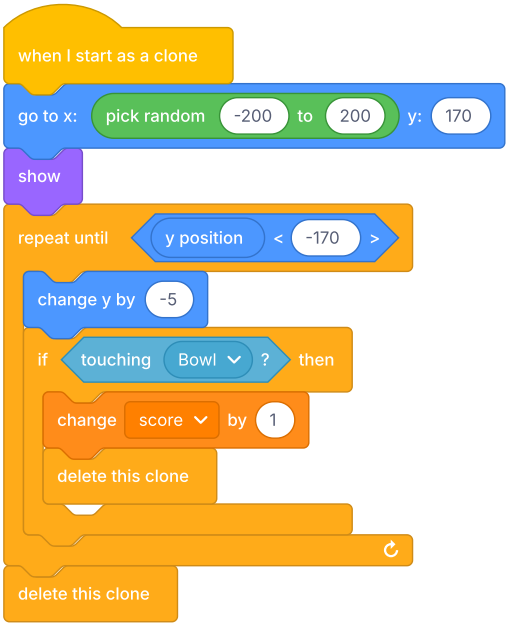
Bowl



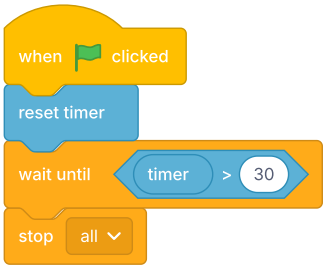


Star





Clock



Answer key — worksheet

1. **Key words:** game balance = Making a game hard but fair. · parameter = A number that changes how the game feels. · lives = Chances left before the game ends.
2. **Which part of the game handles it?:** Player script: Moving with arrow keys · Original (factory): Creating a new star every 0.5 s, Hiding at the start · Clone script: Falling from a random place, Scoring when touching the player
3. **My plan:** A clear plan that covers every requirement.
4. **Checklist:** All requirements met.
5. **Two stars and a wish:** Kind, specific feedback and an improvement.

Kitabash original curriculum. Scratch is a project of the Scratch Foundation in collaboration with the MIT Media Lab; this course is not affiliated with or endorsed by them.